AMIT

AI/ML Engineer | Generative AI & RAG

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SUMMARY

Aspiring AI/ML Engineer and MCA (Software Engineering) graduate with hands-on experience building Machine Learning, Deep Learning, and Generative AI applications, including Retrieval-Augmented Generation (RAG) systems, predictive models, and end-to-end data pipelines in Python. Skilled in Large Language Model (LLM) integration, Natural Language Processing (NLP), vector search, and data analysis, with a strong problem-solving foundation in Object-Oriented Programming (OOP) and Data Structures & Algorithms (200+ problems solved on LeetCode). Eager to apply and grow AI/ML expertise to build production-ready, real-world solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, JavaScript

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Model Evaluation, Deep Learning Fundamentals, Scikit-learn

Generative AI & NLP: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LangChain, Prompt Engineering, Natural Language Processing (NLP), Vector Databases (FAISS), Embeddings

Data Analysis & Visualization: Exploratory Data Analysis (EDA), Data Cleaning, Pandas, NumPy, Matplotlib, Plotly

Backend & APIs: FastAPI, REST APIs, Streamlit

Web Development: HTML, CSS, Tailwind CSS, React

Databases: PostgreSQL, MySQL, SQLite

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

Core CS Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)

PROJECTS

MedInsight AI — Retrieval-Augmented Generation (RAG) Medical Assistant | GitHub | Live Demo

Python, LangChain, FAISS, ONNX Runtime, Groq LLM, Streamlit

- Designed and built a Retrieval-Augmented Generation (RAG) medical assistant that ingests unstructured PDF, XML, and CSV medical records and returns context-grounded, LLM-generated answers with cited source snippets, solving the problem of scattered, hard-to-search medical documentation.
- Engineered the full retrieval pipeline — ONNX-based text embeddings (all-MiniLM-L6-v2), FAISS vector indexing, and similarity-based relevance scoring — combined with prompt engineering to feed a Groq-hosted LLM for accurate, low-latency responses.
- Delivered a production-style Streamlit application with streaming responses, session-based chat history, and chat export, while trimming heavy dependencies to keep the app lightweight for cloud deployment.

Vendor Invoice Intelligence System | GitHub | Live Demo

Python, SQL (SQLite), Scikit-Learn, Streamlit, Plotly

- Built a freight cost prediction model (Linear Regression) achieving an R^2 of 0.963 and RMSE of 124.66, outperforming Random Forest and Decision Tree baselines to automate logistics cost estimation from invoice data.
- Developed a Random Forest classification model with balanced class weights to flag high-risk invoices, joining vendor and purchase SQL tables and engineering 5 predictive features to improve audit efficiency.
- Designed SQL-driven data pipelines and an interactive Streamlit + Plotly dashboard for freight prediction, invoice risk scoring, and model performance comparison, enabling faster, data-backed audit decisions.

EDUCATION

Banaras Hindu University, Uttar Pradesh, India
MCA (Software Engineering) — CGPA: 7.98 — Aug 2024 – Jul 2026

University of Delhi, New Delhi, India
B.Sc. (Hons.) Computer Science — CGPA: 8.20 — Nov 2020 – Jul 2023

CERTIFICATIONS

- **LinkedIn Learning & Microsoft** — Career Essentials in Data Analysis
- **NPTEL (IIT Madras)** — Business Intelligence & Analytics (Elite, 84%)

ACHIEVEMENTS

- Solved 200+ Data Structures & Algorithms problems on LeetCode, building a strong foundation in algorithmic problem-solving and coding efficiency.
- Built and deployed 2 end-to-end AI/ML applications (RAG-based and predictive-modeling based) to production, from data pipeline to live cloud-hosted demo.